



**COMPETENCY STANDARD
FOR
MACHINE MANAGEMENT AND MAINTENANCE
(MECHANICAL) FOR TEXTILE INDUSTRY
TEXTILE SECTOR**

**Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance
People's Republic of Bangladesh**

COPYRIGHT

The Competency Standard for Machine Management and Maintenance (Mechanical) for Textile Industry is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry so that individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

This document has been developed for use in training under the Skills for Industry Competitiveness and Innovation Program (SICIP) to meet the skills requirements of the Textile sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

Public and private institutions may use the information contained in this standard for activities benefitting Bangladesh.

Other interested parties must obtain permission from the owner of this document for reproduction of information in any manner in whole or in part of this Skills Standard, in English or other languages.

This document is available at:

Skills for Industry Competitiveness and Innovation Program (SICIP)

Finance Division, Ministry of Finance

Probashi Kallyan Bhaban (Level-15 & 16)

71-72 Old Elephant Road, Eskaton Garden

Dhaka-1000

Phone: +8802 55138753-55, Fax: +88 02 55138752

Website: <https://sicip.gov.bd/>

INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The textile and apparel sector is the largest manufacturing and export sector of Bangladesh. Modern spinning, weaving, knitting, dyeing-finishing and garment factories depend on complex mechanical systems -- spindles, rollers, gears, cams, belts, chains, bearings, drives and precision-aligned moving parts -- that must run continuously and accurately for extended periods. Mechanical wear, misalignment, inadequate lubrication and untimely breakdowns directly reduce machine uptime, compromise product quality and increase maintenance cost. The industry therefore requires technicians who combine sound mechanical fundamentals with hands-on, textile-specific competence in preventive, predictive and corrective maintenance, fault diagnosis, precision repair and machine alignment, alongside a working awareness of the mechanical drives, sensors and PLC-based control elements increasingly integrated into modern textile machinery.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of the textile industry's maintenance workforce by implementing market-responsive and job-focused training programs through public and private training providers. This requires the development of competency standards for each of the occupations/trades, which provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical sessions, which aligns with modern learning preferences and supports flipped classroom models. Industry visits and industry-based practical sessions in textile mills and their maintenance departments are strongly recommended to expose trainees to actual production and maintenance machinery.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements. The document details the format, sequencing, wording, and layout of the Competency Standard for the occupation, which comprises Units of Competence and their corresponding Elements.

OVERVIEW

A Competency Standard is a written specification of the knowledge, skills, and attitudes required for the effective performance of an occupation or trade role, corresponding to the standards of performance required in the workplace.

A Competency Standard:

- Provides a consistent and reliable set of components for training, recognizing, and assessing workers' skills
- Enables industry-recognized qualifications to be awarded through direct assessment of practical workplace competencies
- Encourages the development and delivery of flexible training that suits individual needs and industry requirements
- Supports learning and assessment in a work-related environment leading to verifiable workplace outcomes

Competency Standards describe the skills, knowledge, and attitudes needed to perform effectively in the workplace. They acknowledge that workers can achieve vocational and technical competency through many pathways, emphasizing what the learner can DO rather than how or where they learned it.

This Competency Standard for Machine Management and Maintenance (Mechanical) for Textile Industry has been developed with the following key considerations:

- Target Trainees: HSC passed or equivalent; candidates holding a Diploma in Mechanical/Textile Engineering; candidates having prior hands-on experience in mechanical fitting, machine maintenance or a related trade, will be given preference.
- Job Roles: Mechanical Maintenance Technician, Machine Fitter/Mechanic, Junior Maintenance Engineer, Workshop Technician and Maintenance Supervisor (entry-level) in spinning, weaving, knitting, dyeing-finishing and garment factories.
- Industry Context: Bangladesh's textile industry including spinning mills, weaving mills, knitting units, wet processing (dyeing, printing and finishing) plants, garment factories and their maintenance/utility sections.
- Course Duration: 360 hours delivered over three months.
- Batch Size: 25 trainees per batch; a total of 300 trainees (12 batches) will be trained under this course.
- Language of Delivery: Training may be delivered in Bangla with English technical terminology; practical demonstration and oral assessment methods are prioritized.
- Delivery Mode: Approximately 30% theory and 70% practical, including industry exposure.

A Unit of Competency describes a distinct, work-related activity normally undertaken by one person in accordance with industry standards. Each unit is documented in a standard format comprising:

- Unit Code, Title, and Descriptor
- Elements of Competency and Performance Criteria
- Range of Variables
- Curricular Content Guide
- Assessment Evidence Guide

Unit codes follow the format SICIP-TEX-MMM-nn-X, where MMM denotes Machine Management and Maintenance (Mechanical) and X denotes G (Generic), S (Sector Specific) or O (Occupation Specific).

UNITS & ELEMENTS AT A GLANCE

Category of Competency	Duration
Generic Competencies	20 Hrs.
Sector Specific Competencies	20 Hrs.
Occupation Specific Competencies	320 Hrs.
Total	360 Hrs.

Generic Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Hours
SICIP-BPI- MMM-01-G	Apply Occupational Health and Safety (OHS) Practices at Textile Worksites	1. Identify OHS policies and procedures 2. Use personal health and safety practices 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-BPI- MMM-02-G	Operate in a Team Environment	1. Identify own role and responsibilities within the team 2. Support and cooperate with team members 3. Communicate and share information within the team 4. Contribute to problem-solving within the team	10
Total Hours			20 Hrs.

Sector Specific Competencies (10 Hrs.)

Code	Unit of Competency	Elements of Competency	Hours
SICIP-TEX- MMM-01-S	Work in the Textile Sector	1. Identify the textile sector and its main activities 2. Understand basic roles in textile production processes 3. Recognize common tools and equipment used 4. Understand simple quality requirements in textile work	10
SICIP-TEX- MMM-02-S	Apply Green Practices in the Textile Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hours			20 Hrs.

Occupation Specific Competencies (320 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX- MMM-01-O	1. Interpret Textile Machinery and Maintenance Work	1. Interpret preventive, predictive and corrective maintenance. 2. Identify types and functions of textile machinery (spinning, weaving, knitting and finishing) 3. Interpret machine manuals, drawings and maintenance schedules 4. Identify hand tools, power tools and maintenance equipment 5. Interpret work orders, job cards and maintenance requisitions 6. Apply workplace safety and PPE requirements for maintenance work	50
SICIP-TEX- MMM-02-O	2. Perform Preventive Maintenance of Textile Machinery	1. Prepare for preventive maintenance activities 2. Carry out scheduled inspection and maintenance as per checklist 3. Identify wear and potential failure points 4. Record maintenance activities and update maintenance history 5. Prepare lubrication schedule and materials 6. Carry out lubrication operations and monitor lubricant condition-consumption	60
SICIP-TEX- MMM-03-O	3. Perform Mechanical Fault Diagnosis and Troubleshooting	1. Find different types of sensors/ tools to identify common mechanical faults in textile machinery 2. Use diagnostic tools and techniques to locate faults 3. Analyze fault causes using problem-solving methods 4. Determine appropriate corrective action 5. Record fault diagnosis and rectification details	50
SICIP-TEX- MMM-04-O	4. Perform Repair, Overhaul and Replacement of Mechanical	1. Prepare for repair/replacement of mechanical parts 2. Dismantle and inspect mechanical components (bearings, gears, belts, chains,	50

	Components	shafts) 3. Repair or replace worn/damaged components 4. Reassemble and test the machine after repair 5. Maintain repair records and update spare parts usage	
SICIP-TEX-MMM-05-O	5. Perform Machine Alignment, Calibration and Installation Support	1. Prepare for machine alignment, calibration and installation tasks 2. Carry out shaft, belt and roller alignment 3. Calibrate machine settings, tension and pressure devices 4. Support installation and commissioning of new or relocated machines	30
SICIP-TEX-MMM-06-O	6. Integrate cross-disciplinary technology	1. Diagnose and repair Integrated Systems 2. Repair mechanical drives 3. Interpret PLC electrical circuits 4. Understand advanced textile material sensors	20
SICIP-TEX-MMM-07-O	7. Perform Basic Welding, Fabrication and Workshop Support Operations	1. Prepare for welding and fabrication tasks 2. Carry out basic welding, cutting and fabrication operations 3. Fabricate or repair simple machine parts, guards and fixtures 4. Maintain workshop tools, machines and housekeeping	30
SICIP-TEX-MMM-08-O	8. Maintain Maintenance Records, Spare Parts Inventory and Safety Compliance	1. Maintain machine history cards and maintenance logs 2. Manage spare parts inventory and requisition process 3. Monitor compliance with safety, environmental and housekeeping standards 4. Report equipment downtime, breakdowns and maintenance KPIs	30
Total Hours			320 Hrs.

COMPETENCY STANDARD: MACHINE MANAGEMENT AND MAINTENANCE FOR TEXTILE INDUSTRY

The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-MMM-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply Occupational Health and Safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures; applying personal health and safety practices; reporting hazards and risks; and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1. 1 <u>OHS policies</u> and safe operating procedures are read and understood. 1. 2 Safety signs and symbols are identified and followed. 1. 3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety practices	2. 1 OHS policies and procedures are followed and practiced. 2. 2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2. 3 Personal hygiene is maintained.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to the authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4. 1 Alarms and warning devices are responded to. 4. 2 <u>Emergency response plans and procedures</u> are implemented. 4. 3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables:

Variable	Range (May include but not limited to)
1. OHS policies	1. 1 International OHS requirements 1. 2 Bangladesh standards for OHS 1. 3 Bangladesh Labour Act and Rules 1. 4 Building Code 1. 5 Fire Safety Rules and Regulations 1. 6 Industry guidelines and mill safety manual
2. Personal Protective Equipment (PPE)	2. 1 Nose/face mask (dust mask) 2. 2 Ear plugs or ear muffs 2. 3 Hand gloves 2. 4 Safety shoes 2. 5 Head cap/helmet 2. 6 Apron and overalls 2. 7 Goggles and safety glasses 2. 8 Hair net
3. Hazards and risks	3.1 Physical hazards (moving machine parts, nip points, noise, vibration, heat) 3.2 Chemical hazards (lubricants, cleaning agents, solvents) 3.3 Biological hazards (bacteria, fungus in damp areas) 3.4 Ergonomic hazards (repetitive work, standing, manual lifting) 3.5 Electrical hazards 3.6 Fire hazards (cotton dust, fly accumulation, static electricity)
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first aid 4.5 Electrical accident response 4.6 Emergency machine stop procedures
5. First aid procedure	5. 1 Washing of open wound 5. 2 Washing chemically infected area 5. 3 Applying bandage 5. 4 Tourniquet 5. 5 Applying CPR (Cardiopulmonary Resuscitation) 5. 6 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures in a textile mill. 1.3 Emergency procedures. 1.3.1 Firefighting 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response.
---------------------------	---

	1.4 Types of hazards (biological, chemical, physical, ergonomic and electrical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 Safety signs, symbols and colour codes. 1.8 OHS awareness and reporting requirements.
2. Underpinning Skills	2. 1 Identifying OHS policies and procedures 2. 2 Following personal work safety practices 2. 3 Selecting and using appropriate PPE 2. 4 Reporting hazards and risks 2. 5 Responding to emergency procedures 2. 6 Maintaining physical well-being in the workplace 2. 7 Performing basic firefighting using fire extinguishers 2. 8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 Safety signs and symbols 4.6 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Reported incidents arising from hazards and risks to the authority. 1.4 Implemented emergency response plans and procedures. 1.5 Applied basic first aid procedure.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
---------------------------------	--

Unit of Competency: WORK IN A TEAM ENVIRONMENT	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-MMM-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within the team and with other workers are identified.
2. Communicate and cooperate with team members	2. 1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2. 2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2. 3 Diversity in character is respected and valued in team functioning. 2. 4 Views and opinions of other team members are understood and valued. 2. 5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with the team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications, and workplace procedures. 3.3 Support to other team members is provided to ensure the team achieves its goals. 3.4 Agreed reporting lines are followed using standard operating procedures.
4. Solve problems as a team member	4.1 Current and potential problems faced by the team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables:

Variable	Range (May include but not limited to)
1. Forms of communication	1. 1 Agenda 1. 2 Simple reports such as progress and incident reports 1. 3 Job sheets and shift handover log 1. 4 Operational manuals and SOPs 1. 5 Brochures and promotional material 1. 6 Visual and graphic materials 1. 7 Standards 1. 8 OSH information 1. 9 Signs and symbols

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Team goals and collaborative decision-making processes 1. 2 Roles and responsibilities of team members 1. 3 Relationships within the team and with other workers 1. 4 Effective interpersonal skills to interact with team members 1. 5 Effective formal and informal forms of communication 1. 6 Value of diversity in team functioning 1. 7 Correct use of workplace terminology 1. 8 Team's duties, responsibilities, authorities, objectives and task requirements 1. 9 Support mechanisms for other members of the team to ensure achievement of goals 1. 10 Methods of identifying current and potential problems faced by a team 1. 11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2. 1 Identifying team goals and collaborative decision-making processes 2. 2 Identifying roles and responsibilities of team members 2. 3 Using effective interpersonal skills to interact with team members 2. 4 Using formal and informal forms of communication 2. 5 Understanding and valuing views and opinions of other team members 2. 6 Performing tasks in accordance with organizational and team requirements, specifications and workplace procedures 2. 7 Supporting other members of the team to ensure the team achieves its goals 2. 8 Identifying current and potential problems faced by the team 2. 9 Identifying solutions to problems 2. 10 Solving problems effectively and evaluating the outcome of the implemented solution

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals/Guidelines 4.3 Pens 4.4 Papers 4.5 Workbooks

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified team goals and work processes. 1. 2 Communicated and cooperated with team members. 1. 3 Worked as a team member. 1. 4 Solved problems as a team member.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 4. Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: WORK IN THE TEXTILE SECTOR	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-MMM-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for work in the textile sector. It specifically includes the tasks of identifying the textile sector and its main activities, understanding basic roles in textile production processes, recognizing common tools and equipment used, and understanding simple quality requirements in textile work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the textile sector and its main activities	1. 1 The textile sector is identified within the context of national and global industries. 1. 2 <u>Main activities</u> of the textile sector are recognized. 1. 3 <u>Different stages of the textile value chain</u> are described in simple terms. 1. 4 Key workplaces in the textile sector are identified. 1. 5 Importance of the textile sector in the economy and the environment is understood.
2. Understand basic roles in textile production processes	2. 1 <u>Basic job roles in textile production</u> are identified and described. 2. 2 Simple production workflows and task sequences are understood. 2. 3 Teamwork and coordination in production processes are identified. 2. 4 Workplace safety roles and responsibilities are understood and followed. 2. 5 Shift working practices and reporting lines are understood.
3. Recognize common tools and equipment used	3.1 <u>Common tools used in textile processing</u> are identified correctly. 3.2 <u>Basic machines and equipment used in production</u> are recognized. 3.3 Functions of tools and equipment are understood in simple terms. 3.4 Basic maintenance and cleaning needs of tools and equipment are recognized.
4. Understand simple quality requirements in textile work	4.1 Basic quality standards in textile work are identified. 4.2 <u>Simple defects in textile products</u> are recognized. 4.3 Importance of maintaining quality during production is understood. 4.4 Work output is checked against basic quality

Elements of Competency	Performance Criteria
	expectations. 4.5 Good workmanship practices are followed to ensure quality.

Range of Variables:

Variable	Range (May include but not limited to)
1. Main activities	1. 1 Fibre production and preparation 1. 2 Yarn manufacturing (spinning) 1. 3 Fabric manufacturing (weaving and knitting) 1. 4 Wet processing (dyeing, printing and finishing) 1. 5 Garment manufacturing 1. 6 Testing and quality assurance 1. 7 Packaging, storage and export
2. Different stages of the textile value chain	2. 1 Natural and man-made fibre production 2. 2 Fibre preparation and mixing 2. 3 Yarn manufacturing (spinning) 2. 4 Fabric manufacturing (weaving/knitting) 2. 5 Dyeing, printing and finishing 2. 6 Cutting, sewing and finishing of garments 2. 7 Packaging, distribution and export
3. Basic job roles in textile production	3.1 Blow room operator/helper 3.2 Card tenter 3.3 Draw frame and comber operator 3.4 Simplex (speed frame) operator 3.5 Ring frame sider and doffer 3.6 Winding and packing operator 3.7 Quality checker/laboratory assistant 3.8 Maintenance fitter/electrician 3.9 Line leader/supervisor
4. Common tools used in textile processing	4.1 Sliver and roving cutter, scissors and nippers 4.2 Hooks, brushes and dusting tools 4.3 Spanner, screwdriver and plier set 4.4 Oil can and grease gun 4.5 Feeler and setting gauges 4.6 Wrap reel, wrap block and electronic balance 4.7 Tachometer and hand lens 4.8 Vacuum cleaner and air blowing gun
5. Basic machines and equipment used in production	5. 1 Bale opener/plucker and blow room line 5. 2 Carding machine 5. 3 Draw frame (breaker and finisher) 5. 4 Lap former and comber 5. 5 Speed frame (simplex) 5. 6 Ring frame 5. 7 Winding machine/autoconer

Variable	Range (May include but not limited to)
	5. 8 Humidification and waste collection plant 5. 9 Testing laboratory equipment
6. Simple defects in textile products	6. 1 Neps and slubs 6. 2 Thick and thin places 6. 3 Count variation and uneven yarn 6. 4 High hairiness 6. 5 Foreign fibre and contamination 6. 6 Sliver and roving breakage 6. 7 Soft/hard and damaged packages 6. 8 Oil and dirt stains

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Structure and importance of the textile sector in the national and global economy. 1. 2 Main activities and stages of the textile value chain. 1. 3 Position of yarn manufacturing within the textile value chain. 1. 4 Basic job roles, responsibilities and reporting lines in textile production. 1. 5 Simple production workflows, task sequences and shift practices. 1. 6 Common tools, machines and equipment used in textile production and their functions. 1. 7 Basic maintenance and cleaning needs of tools and equipment. 1. 8 Basic quality standards and common defects in textile products. 1. 9 Importance of quality, productivity and good workmanship in the workplace.
2. Underpinning Skills	2. 1 Identifying the textile sector and its main activities 2. 2 Describing the stages of the textile value chain in simple terms 2. 3 Understanding basic job roles and responsibilities in textile production processes 2. 4 Following simple production workflows, task sequences and teamwork practices 2. 5 Applying workplace safety roles and responsibilities in production activities 2. 6 Recognizing common tools, machines and equipment and their functions 2. 7 Recognizing basic maintenance and cleaning needs of tools and equipment 2. 8 Recognizing simple defects and checking work output against basic quality requirements 2. 9 Following good workmanship practices to ensure

	consistent quality output
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Charts, models and samples of fibre, sliver, roving and yarn 4.3 Multimedia projector, laptop and audio-visual clips 4.4 Learning manual and handouts 4.5 Stationery

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified the textile sector and its main activities. 1. 2 Understood basic roles in textile production processes. 1. 3 Recognized common tools and equipment used. 1. 4 Understood simple quality requirements in textile work.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN THE TEXTILE SECTOR	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-MMM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in the textile sector. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1. 1 Basic concepts of environmental sustainability and green practices are explained clearly. 1. 2 <u>Environmental impacts of textile production</u> processes are identified. 1. 3 <u>Resource efficiency concepts</u> are interpreted. 1. 4 Waste reduction, reuse and recycling principles are described in relation to yarn manufacturing operations. 1. 5 <u>Eco-friendly practices in the textile sector</u> are recognized and related to workplace activities.
2. Minimize resource use in the workplace	2. 1 Electricity, water, compressed air and raw materials are used efficiently according to workplace requirements. 2. 2 <u>Unnecessary consumption of resources</u> is identified and avoided during operations. 2. 3 Machines and equipment are operated in energy-efficient modes where applicable. 2. 4 Material wastage is minimized through careful handling and accurate measurement. 2. 5 <u>Resource-saving practices</u> are applied where possible. 2. 6 Leakage, spillage and wastage of materials are promptly controlled and reported. 2. 7 <u>Workplace practices</u> for resource conservation are followed.
3. Implement waste management practices	3.1 <u>Types of waste</u> generated in yarn manufacturing operations are identified correctly. 3.2 Waste materials are segregated according to workplace and environmental requirements. 3.3 <u>Reusable and recyclable materials</u> are separated from non-recyclable waste. 3.4 Waste is stored safely in designated containers or areas. 3.5 Waste disposal is carried out following environmental rules and organizational procedures.

Elements of Competency	Performance Criteria
	3.6 Hazardous waste is handled carefully according to safety guidelines. 3.7 Waste reduction practices are applied to minimize environmental impact during operations.

Range of Variables:

Variable	Range (May include but not limited to)
1. Environmental impacts of textile production	1. 1 High electrical energy consumption of spinning machinery and humidification plants 1. 2 Cotton dust, fibre fly and micro-fibre emission 1. 3 Noise pollution from machines and compressors 1. 4 Solid waste generation (droppings, comber noil, soft and hard waste) 1. 5 Water consumption and effluent from wet processing 1. 6 Air emission and greenhouse gases from boilers and generators 1. 7 Used oil, grease and contaminated cotton waste
2. Resource efficiency concepts	2. 1 Energy-efficient motors and inverter/variable frequency drives 2. 2 Control of compressed air leakage and optimized air pressure 2. 3 Optimized humidification and ventilation control 2. 4 Reduction of soft and hard waste percentage 2. 5 Preventive maintenance to reduce power consumption 2. 6 Energy-efficient lighting and switching-off practices 2. 7 Optimum machine loading and reduction of idle running
3. Eco-friendly practices in the textile sector	3.1 Recycling of soft waste back into mixing where permitted 3.2 Use of recycled, organic or sustainably sourced fibre 3.3 Dust collection, filter and waste recovery systems 3.4 Compliance with Department of Environment (DoE) rules and regulations 3.5 Effluent treatment plant (ETP) and rainwater harvesting 3.6 Reuse of cans, bobbins, cones, cartons and packaging materials 3.7 Green building and energy audit practices
4. Unnecessary consumption of resources	4.1 Idle running of machines and motors 4.2 Compressed air leakage in pipelines and guns 4.3 Excess lighting in unoccupied areas 4.4 Over-cleaning with air guns instead of vacuum 4.5 Excess lubricant and cleaning material use 4.6 Excessive good fibre loss with droppings and noil
5. Resource-saving practices	5. 1 Switching off machines, lights and fans when not required

Variable	Range (May include but not limited to)
	5. 2 Reporting air, water and oil leakage promptly 5. 3 Using vacuum cleaning instead of air blowing where possible 5. 4 Reusing cans, tubes, cones and packaging materials 5. 5 Maintaining correct machine settings to reduce waste 5. 6 Recording and monitoring waste and energy use
6. Workplace practices	6. 1 Standard operating procedures (SOPs) 6. 2 Environmental policy of the organization 6. 3 Housekeeping and 5S practices 6. 4 Energy and waste monitoring records 6. 5 Instructions from supervisors and safety notices
7. Types of waste	7.1 Blow room droppings and card droppings/flat strips 7.2 Comber noil 7.3 Sliver, roving and pneumafil soft waste 7.4 Hard waste from winding (yarn and thread waste) 7.5 Damaged cans, bobbins, cones and tubes 7.6 Packaging waste (poly bags, bale wrappers, straps, cartons) 7.7 Used oil, grease and oil-contaminated cotton (hazardous waste) 7.8 Used filters, sweeps and floor waste
8. Reusable and recyclable materials	8.1 Usable soft waste for back-mixing 8.2 Comber noil for coarse count and open-end spinning 8.3 Cans, bobbins, tubes and cones 8.4 Cartons, poly bags and bale straps 8.5 Metal and plastic scrap 8.6 Paper and stationery waste

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Basic concepts of environmental sustainability, green practices and climate-resilient production. 1. 2 Environmental impacts of textile and yarn manufacturing operations. 1. 3 Resource efficiency concepts relating to energy, water, compressed air and raw material. 1. 4 Principles of waste reduction, reuse and recycling in a spinning mill. 1. 5 Eco-friendly practices and technologies applied in the textile sector. 1. 6 Types of waste generated at each stage of yarn manufacturing and permissible waste levels. 1. 7 Waste segregation, storage, disposal methods and compliance with environmental regulations. 1. 8 Hazardous waste handling and workplace environmental safety procedures.
---------------------------	--

	1. 9 Relevant national environmental rules and organizational environmental policy.
2. Underpinning Skills	2. 1 Interpreting green concepts and relating them to workplace activities 2. 2 Identifying environmental impacts of yarn manufacturing operations 2. 3 Applying resource efficiency principles in workplace operations 2. 4 Using electricity, compressed air, water and raw materials efficiently 2. 5 Identifying and avoiding unnecessary consumption of resources 2. 6 Operating machines and equipment in energy-efficient modes where applicable 2. 7 Handling materials carefully to reduce wastage and improve accuracy 2. 8 Controlling and reporting leakage, spillage and resource wastage promptly 2. 9 Identifying and segregating different types of waste generated in yarn manufacturing 2. 10 Separating reusable and recyclable materials from non-recyclable waste 2. 11 Handling hazardous waste according to safety guidelines and regulations
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Standard operating procedures 4. 3 Workplace documents, signs and symbols 4. 4 Codes of conduct and environmental policy 4. 5 Waste segregation bins and storage area 4. 6 Tools, equipment and facilities appropriate to the process or activities 4. 7 Materials relevant to the proposed activity

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Interpreted green concepts.
-----------------------------------	--

	1. 2 Minimized resource use in the workplace. 1. 3 Implemented waste management practices.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: INTERPRET TEXTILE MACHINERY AND MAINTENANCE WORK	Nominal Duration: 50 hrs.	Unit Code: SICIP-TEX-MMM-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret textile machinery and maintenance work. It specifically includes the tasks of interpreting preventive, predictive and corrective maintenance, identifying types and functions of textile machinery, interpreting machine manuals, drawings and maintenance schedules, identifying hand tools, power tools and maintenance equipment, interpreting work orders, job cards and maintenance requisitions, and applying workplace safety and PPE requirements for maintenance work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret preventive, predictive and corrective maintenance	1.1 <u>Types of maintenance strategies</u> (preventive, predictive and corrective) are differentiated according to industry practice. 1.2 Purpose and scope of each maintenance strategy are explained in relation to machine uptime and cost. 1.3 Maintenance strategy applicable to a given machine or situation is identified according to organizational policy. 1.4 Basic terms used in maintenance management (breakdown, downtime, MTBF, MTTR) are interpreted.
2. Identify types and functions of textile machinery	2.1 <u>Types of textile machinery</u> are identified according to the stage of textile manufacturing. 2.2 Function of each machine type in the production process is explained. 2.3 Major <u>mechanical assemblies/sub-systems/components</u> of common textile machines are identified. 2.4 Machine nameplate and specification details are read and interpreted.
3. Interpret machine manuals, drawings and maintenance schedules	3.1 <u>Machine manuals and technical documents</u> are identified and interpreted to obtain required maintenance information. 3.2 <u>Mechanical drawings</u> are read to identify parts, dimensions and assembly sequence. 3.3 Maintenance schedules and preventive maintenance charts are interpreted according to machine requirement. 3.4 Information gaps or inconsistencies in documents are identified and reported to the supervisor.
4. Identify hand tools, power tools and maintenance equipment	4.1 <u>Hand tools</u> used in mechanical maintenance are identified according to their function. 4.2 <u>Power tools and maintenance equipment</u> are identified and selected according to task requirement. 4.3 Tools and equipment are checked for serviceability and

	safety before use. 4.4 Tools are used, cleaned and stored according to standard procedure.
5. Interpret work orders, job cards and maintenance requisitions	5.1 Work order/job card information is interpreted to determine scope, priority and resources required for the task. 5.2 Maintenance requisition/spare parts request is prepared according to job requirement. 5.3 Work order status is updated at each stage of job completion. 5.4 Job card is closed and submitted for approval following completion of work.
6. Apply workplace safety and PPE requirements for maintenance work	6.1 Personal protective equipment (PPE) required for the maintenance task is identified and worn correctly. 6.2 Workplace safety procedures (lockout-tagout, permit-to-work) are followed before starting maintenance work. 6.3 Hazards associated with the maintenance task are identified and controlled. 6.4 Housekeeping and 5S practices are maintained at the work area.

Range of Variables:

Variables	Range (may include but not limited to)
1. Types of maintenance strategies	1.1 Preventive maintenance 1.2 Predictive maintenance 1.3 Corrective/breakdown maintenance
2. Types of textile machinery	2.1 Spinning machines (blow room, carding, drawing, roving, ring frame) 2.2 Weaving machines (looms) 2.3 Knitting machines (circular, flat bed) 2.4 Finishing/utility machines
3. Mechanical assemblies/ sub-systems/ components	3.1 Boiler 3.2 Compressor 3.3 Heat Exchanger 3.4 Generator 3.5 Gear box 3.6 Pneumatic cylinder and valve 3.7 Pump 3.8 Blower 3.9 Bearing 3.10 Filter, seal, valve
4. Machine manuals and technical documents	4.1 Operation manual 4.2 Maintenance manual 4.3 Spare parts catalogue 4.4 OEM technical bulletin
5. Mechanical drawings	5.1 Assembly drawing 5.2 Part drawing 5.3 Exploded view 5.4 Bill of materials 5.5 Circuit diagram (Electrical, Pneumatic, Hydraulic)
6. Hand tools	6.1 Spanners and wrenches

	6.2 Screwdrivers 6.3 Pliers and cutters 6.4 Hammers and punches 6.5 Measuring tools (vernier, feeler gauge)
7. Power tools and maintenance equipment	7.1 Drill machine 7.2 Grinder 7.3 Hydraulic/pneumatic puller 7.4 Hoist/crane 7.5 Lifting jack
8. Hazards	8.1 Fire/Explosion 8.2 Air/Oil leakage 8.3 Electrical Shock 8.4 Toxic chemical 8.5 Noise and vibration 8.6 High temperature 8.7 Lockout-tagout (LOTO) failure

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Concept and purpose of preventive, predictive and corrective maintenance 1.2 Types, functions and main assemblies of textile machinery 1.3 Machine manuals, drawings and maintenance documentation 1.4 Common hand tools, power tools and maintenance equipment 1.5 Work order/job card and maintenance requisition procedure 1.6 Workplace safety procedures relevant to maintenance work 1.7 Basic maintenance management terms (breakdown, downtime, MTBF, MTTR) 1.8 Machine nameplate label (voltage, current, torque, power, efficiency, etc.) and specification interpretation 1.9 Reporting procedure for information gaps or documentation inconsistencies
2. Underpinning Skills	2.1 Differentiating preventive, predictive and corrective maintenance situations 2.2 Identifying types and functions of textile machinery 2.3 Reading machine manuals, drawings and maintenance schedules 2.4 Selecting and using hand tools, power tools and maintenance equipment safely 2.5 Interpreting and completing work orders and job cards 2.6 Applying PPE and workplace safety procedures during maintenance work 2.7 Reading machine nameplates and technical specifications 2.8 Identifying components of textile machineries (mechanical/electrical) 2.9 Identifying and reporting hazards at the maintenance work area
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety

	3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Sample textile machines/sub-assemblies or models (spinning, weaving, knitting) 4.3 Machine manuals, drawings and maintenance schedules 4.4 Hand tools, power tools and maintenance equipment 4.5 PPE set (helmet, safety shoes, gloves, eye/ear protection) 4.6 Sample work orders, job cards and requisition forms 4.7 Maintenance management software/CMMS demo or register format 4.8 Learning manual and projector

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 interpreted preventive, predictive and corrective maintenance 1.2 identified types and functions of textile machinery 1.3 interpreted machine manuals, drawings and maintenance schedules 1.4 identified hand tools, power tools and maintenance equipment 1.5 interpreted work orders, job cards and maintenance requisitions 1.6 applied workplace safety and PPE requirements for maintenance work
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated textile mill workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM PREVENTIVE MAINTENANCE OF TEXTILE MACHINERY	Nominal Duration: 60 hrs.	Unit Code: SICIP-TEX-MMM-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform preventive maintenance of textile machinery. It specifically includes the tasks of preparing for preventive maintenance activities, carrying out scheduled inspection and maintenance as per checklist, identifying wear and potential failure points, recording maintenance activities and updating maintenance history, preparing lubrication schedule and materials, and carrying out lubrication operations and monitoring lubricant condition-consumption.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for preventive maintenance activities	1.1 Preventive maintenance checklist for the assigned machine is obtained and reviewed. 1.2 <u>Tools, spares and consumables</u> required for the task are collected and checked. 1.3 Machine is prepared for maintenance (stopped, isolated, tagged) following safety procedure. 1.4 PPE and work area are prepared according to task requirement.
2. Carry out inspection and maintenance as per checklist	2.1 <u>Machine components</u> are inspected systematically following the <u>preventive maintenance checklist</u> . 2.2 Major components of the machine are checked and ad Fasteners, guards, belts and safety devices are checked and tightened/adjusted as required. 2.3 Cleaning, adjustment and minor servicing tasks are carried out according to checklist items. 2.4 Abnormal conditions observed during inspection are noted for further action.
3. Identify wear and potential failure points	3.1 <u>Signs of wear and deterioration</u> (looseness, noise, vibration, heat, abnormal wear marks) are identified during inspection. 3.2 Components approaching the end of service life are identified based on condition and running hours. 3.3 Potential failure points are prioritized according to criticality and risk. 3.4 Findings are reported to the supervisor for planning corrective/repair action.
4. Record maintenance activities and update maintenance history	4.1 Maintenance activities carried out are recorded in the <u>maintenance record/history card</u> . 4.2 Parts replaced, consumables used and time taken are recorded accurately. 4.3 Maintenance records are updated in the manual/digital system according to organizational procedure.

	4.4 Records are filed/stored for future reference and audit.
5. Carry out machine maintenance	5.1 Machine components are lubricated. 5.2 Failed components are replaced 5.3 Filter and strainer are cleaned 5.4 Pneumatic and Hydraulic systems are inspected and repaired 5.5 Machine is tested and trial run is performed. 5.6 Maintenance activities are documented.

Range of Variables:

Variables	Range (may include but not limited to)
1. Tools, spares and consumables	1.1 Hand tools 1.2 Cleaning materials 1.3 Lubricants and greases 1.4 Common spare parts (bearings, belts, filters)
2. Machine components	2.1 Bearings 2.2 Gearboxes 2.3 Valves, seals, gaskets, 2.4 Chains and sprockets 2.5 Slideways/guideways 2.6 5.5 Cam and linkage points
3. Preventive maintenance checklist	3.1 Daily checklist 3.2 Weekly checklist 3.3 Monthly/periodic checklist
4. Signs of wear and deterioration	4.1 Abnormal noise 4.2 Excessive vibration 4.3 Overheating 4.4 Loose fasteners 4.5 Worn belts/bearings
5. Maintenance record/history card	5.1 Machine history card 5.2 Daily maintenance log 5.3 Digital CMMS record

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Preventive maintenance planning and checklist-based inspection 1.2 Signs of wear, deterioration and impending failure 1.3 Maintenance recording and history-card procedures 1.4 Lubrication points, lubricant types and lubrication schedules 1.5 Basic condition of lubricants and contamination indicators 1.6 Common preventive maintenance tasks (cleaning, tightening, adjustment), alignment, balancing 1.7 Safe isolation procedures before starting maintenance (lockout-tagout) 1.9 Digital/manual maintenance record-keeping systems (CMMS)
2. Underpinning Skills	2.1 Preparing for and carrying out preventive maintenance activities 2.2 Inspecting machine components systematically following a checklist

	2.3 Identifying wear and potential failure points 2.4 Recording maintenance activities and updating maintenance history 2.6 Carrying out lubrication operations monitoring machine noise and vibration 2.7 Isolating and tagging machines safely before maintenance 2.8 Prioritizing potential failure points by criticality and risk
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Sample textile machine(s) requiring preventive maintenance 4.3 Preventive maintenance checklists and machine history cards 4.4 Lubrication chart, lubricants and lubrication tools (grease gun, oil can) 4.5 Hand tools and basic maintenance equipment 4.6 Lockout-tagout kit 4.7 Waste/lubricant disposal containers 4.8 PPE set 4.9 Learning manual and projector

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 prepared for preventive maintenance activities 1.2 carried out scheduled inspection and maintenance as per checklist 1.3 identified wear and potential failure points 1.4 recorded maintenance activities and updated maintenance history 1.5 prepared lubrication schedule and materials 1.6 carried out lubrication operations and monitored lubricant condition-consumption
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of	3.1 Competency assessment must be done in an assessment/

Assessment	training center or in an actual or simulated textile mill workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
------------	---

Unit of Competency: PERFORM MECHANICAL FAULT DIAGNOSIS AND TROUBLESHOOTING	Nominal Duration: 50 hrs.	Unit Code: SICIP-TEX-MMM-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform mechanical fault diagnosis and troubleshooting. It specifically includes the tasks of finding different types of sensors/tools to identify common mechanical faults in textile machinery, using diagnostic tools and techniques to locate faults, analyzing fault causes using problem-solving methods, determining appropriate corrective action, and recording fault diagnosis and rectification details.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Find different types of sensors/tools to identify common mechanical faults in textile machinery	1.1 <u>Common mechanical faults</u> in textile machinery are identified from symptoms (noise, vibration, heat, abnormal output). 1.2 <u>Sensors and simple diagnostic tools</u> used to detect faults are identified. 1.3 Appropriate sensor/tool is selected according to the type of fault suspected. 1.4 Baseline/normal operating condition of the machine is identified for comparison.
2. Use diagnostic tools and techniques to locate faults	2.1 Diagnostic tools are used correctly and safely to check machine condition. 2.2 <u>Diagnostic techniques</u> are applied systematically to locate the fault. 2.3 Fault location is narrowed down to the specific component/assembly. 2.4 Readings and observations are recorded during diagnosis.
3. Analyze fault causes using problem-solving methods	3.1 Possible causes of the identified fault are listed based on machine knowledge and symptoms. 3.2 <u>Problem-solving techniques</u> are applied to identify the root cause. 3.3 Root cause is verified through further checking/testing where necessary. 3.4 Findings are discussed with the supervisor/team before deciding on action.
4. Determine appropriate corrective action	4.1 Corrective action is determined based on the identified root cause and available resources. 4.2 Repair, adjustment or replacement is selected as the appropriate corrective measure. 4.3 Priority and timeline for corrective action are determined based on machine criticality. 4.4 Preventive measures to avoid recurrence are recommended.

5. Record fault diagnosis and rectification details	5.1 Fault symptoms, diagnosis process and root cause are recorded in the <u>fault report/log</u>. 5.2 Corrective action taken and parts used are recorded accurately. 5.3 Downtime and repair time are recorded for performance analysis. 5.4 Recurring fault patterns are flagged for further investigation.
---	--

Range of Variables:

Variables	Range (may include but not limited to)
1. Common mechanical faults	1.1 Bearing failure 1.2 Belt slippage/wear 1.3 Gear wear/backlash 1.4 Misalignment 1.5 Loose coupling 1.6 Excessive vibration
2. Sensors and simple diagnostic tools	2.1 Vibration pen/meter 2.2 Infrared thermometer/thermal camera 2.3 Stethoscope/listening rod 2.4 Tachometer 2.5 Clearance/feeler gauge
3. Diagnostic techniques	3.1 Visual inspection 3.2 Auditory check 3.3 Tactile check (feel for heat/vibration) 3.4 Measurement and comparison with standard
4. Problem-solving techniques	4.1 Fishbone (cause-and-effect) diagram 4.2 5-Why analysis 4.3 Checklist-based troubleshooting
5. Fault report/log	5.1 Fault description 5.2 Root cause 5.3 Corrective action 5.4 Downtime record

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Common mechanical faults and their symptoms in textile machinery 1.2 Sensors and simple diagnostic tools used for fault detection 1.3 Systematic fault-finding and diagnostic techniques 1.4 Root-cause analysis methods (fishbone diagram, 5-Why) 1.5 Fault recording and reporting procedures 1.6 Baseline/normal operating condition of machinery 1.7 Prioritizing corrective actions based on machine criticality 1.8 Recurring fault pattern analysis and preventive recommendations
2. Underpinning Skills	2.1 Identifying common mechanical faults from symptoms 2.2 Selecting and using sensors/diagnostic tools to locate faults 2.3 Applying diagnostic techniques systematically

	2.4 Analyzing fault causes using problem-solving methods 2.5 Determining and justifying appropriate corrective action 2.6 Recording fault diagnosis and rectification details 2.7 Comparing machine readings against baseline/normal condition 2.8 Verifying root-cause findings through further checking/testing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Sample textile machine(s) with simulated/actual faults 4.3 Diagnostic tools (vibration metre, IR thermometer, noise sensor, tachometer, measuring instrument) 4.4 Fault report/log formats 4.5 Fishbone diagram/5-Why analysis templates 4.6 Reference/baseline machine condition data sheets 4.7 PPE set 4.8 Learning manual and projector

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 identified sensors/tools used to detect common mechanical faults in textile machinery 1.2 used diagnostic tools and techniques to locate faults 1.3 analyzed fault causes using problem-solving methods 1.4 determined appropriate corrective action 1.5 recorded fault diagnosis and rectification details
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated textile mill workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM REPAIR, OVERHAUL AND REPLACEMENT OF MECHANICAL COMPONENTS	Nominal Duration: 50 hrs.	Unit Code: SICIP-TEX-MMM-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform repair, overhaul and replacement of mechanical components. It specifically includes the tasks of preparing for repair/replacement of mechanical parts, dismantling and inspecting mechanical components, repairing or replacing worn/damaged components, reassembling and testing the machine after repair, and maintaining repair records and updating spare parts usage.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for repair/ replacement of mechanical parts	1.1 Scope of repair/replacement work is confirmed from the job card/work order. 1.2 <u>Spare parts and materials</u> required are identified, requisitioned and collected. 1.3 Tools and lifting/handling equipment required for the task are prepared. 1.4 Machine is isolated and made safe following lockout-tagout procedure.
2. Dismantle and inspect mechanical components	2.1 Components are dismantled following correct sequence and technique to avoid damage. 2.2 <u>Mechanical components</u> are inspected for wear, damage and dimensional accuracy. 2.3 Removed parts are tagged, arranged and stored to avoid mix-up or loss. 2.4 Inspection findings are compared against manufacturer tolerance/specification.
3. Repair or replace worn/damaged components	3.1 Decision to repair or replace is made based on inspection findings and cost-effectiveness. 3.2 Worn/damaged components are repaired using <u>appropriate methods</u> . 3.3 Components beyond repair are replaced with correct spare parts as per specification. 3.4 Repaired/replaced components are checked for correct fit and function before reassembly.
4. Reassemble and test the machine after repair	4.1 Components are reassembled following correct sequence, torque and fitment procedure. 4.2 Reassembled machine/assembly is checked for correct alignment, clearance and free movement. 4.3 Machine is tested (no-load and load trial) to confirm proper functioning after repair. 4.4 <u>Abnormalities</u> observed during trial run are rectified before handover.

5. Maintain repair records and update spare parts usage	5.1 Repair details, parts replaced and time taken are recorded in the repair/maintenance record. 5.2 Spare parts consumption is recorded and updated in the inventory system. 5.3 Machine history card is updated with the repair/overhaul details. 5.4 Repair report is submitted to the supervisor for review and sign-off.
---	---

Range of Variables:

Variables	Range (may include but not limited to)
1. Spare parts and materials	1.1 Bearings 1.2 Gears 1.3 Belts and chains 1.4 Shafts and couplings 1.5 Seals and gaskets 1.6 Fasteners
2. Mechanical components	2.1 Bearings 2.2 Gears 2.3 Belts 2.4 Chains 2.5 Shafts 2.6 Couplings 2.7 Bushes
3. Appropriate methods	3.1 Machining 3.2 Welding 3.3 Brazing 3.4 Grinding

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Repair and replacement planning for mechanical components 1.2 Dismantling, inspection and tolerance-checking procedures 1.3 Repair methods (machining, welding, resizing) and their applicability 1.4 Reassembly sequence, torque and fitment procedures 1.5 Repair recording and spare parts usage updating 1.6 Lockout-tagout and safe isolation procedures before repair work 1.7 Spare parts identification and requisition procedures 1.8 Trial run and testing procedures after repair
2. Underpinning Skills	2.1 Preparing for repair/replacement of mechanical parts 2.2 Dismantling and inspecting mechanical components against specification 2.3 Repairing or replacing worn/damaged components 2.4 Reassembling and testing machines after repair 2.5 Maintaining repair records and updating spare parts usage 2.6 Isolating and making machines safe before repair work

	2.7 Selecting and using lifting/handling equipment safely 2.8 Checking fit and function of repaired/replaced components
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Sample textile machine/assembly for repair practice 4.3 Mechanical components for dismantling/repair practice (bearings, gears, belts, chains, shafts) 4.4 Hand tools, pullers, lifting/handling equipment 4.5 Measuring instruments (vernier caliper, micrometer, feeler gauge) 4.6 Repair record/spare parts usage formats 4.7 Lockout-tagout kit 4.8 PPE set 4.9 Learning manual and projector

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 prepared for repair/replacement of mechanical parts 1.2 dismantled and inspected mechanical components 1.3 repaired or replaced worn/damaged components 1.4 reassembled and tested the machine after repair 1.5 maintained repair records and updated spare parts usage
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated textile mill workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM MACHINE ALIGNMENT, CALIBRATION AND INSTALLATION SUPPORT	Nominal Duration: 30 hrs.	Unit Code: SICIP-TEX-MMM-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform machine alignment, calibration and installation support. It specifically includes the tasks of preparing for machine alignment, calibration and installation tasks, carrying out shaft, belt and roller alignment, calibrating machine settings, tension and pressure devices, and supporting installation and commissioning of new or relocated machines.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for machine alignment, calibration and installation tasks	1.1 Scope of alignment/calibration/installation task is confirmed from work order/drawing. 1.2 <u>Alignment and calibration tools</u> are selected and checked. 1.3 Machine/foundation condition is checked and prepared prior to the task. 1.4 Safety precautions for the task are identified and implemented.
2. Carry out shaft, belt and roller alignment	2.1 <u>Misalignment types</u> are identified. 2.2 <u>Shaft alignment</u> is carried out using <u>appropriate method</u> and checked within tolerance. 2.3 Belt alignment and tension are checked and adjusted to specification. 2.4 Roller/cylinder alignment (parallelism, level) is checked and corrected as required. 2.5 Alignment results are verified through trial run and measurement.
3. Calibrate machine settings, tension and pressure devices	3.1 <u>Machine settings</u> are checked against specification. 3.2 Tension and pressure devices are calibrated using standard reference/instrument. 3.3 Calibration results are recorded and compared with acceptable tolerance. 3.4 Deviations found during calibration are adjusted or reported for correction.
4. Support installation and commissioning of new or relocated machines	4.1 Installation drawing/layout is interpreted to position the machine correctly. 4.2 Machine is installed/relocated following manufacturer instruction and safety procedure. 4.3 Mechanical connections (foundation, coupling, guarding) are completed and checked. 4.4 Machine is supported through trial run and commissioning checks with the installation team.

Range of Variables:

Variables	Range (may include but not limited to)
1. Alignment and calibration tools	1.1 Dial gauge 1.2 Straight edge 1.3 Feeler gauge 1.4 Spirit level 1.5 Laser alignment tool 1.6 Tension meter
2. Misalignment types	2.1 Horizontal 2.2 Vertical 2.3 Angular
3. Shaft alignment	3.1 Coupling alignment 3.2 Pulley/sheave alignment 3.3 Belt alignment
4. Appropriate methods	4.1 Feeler gauge 4.2 Dial indicator 4.3 Laser alignment
5. Machine settings	5.1 Gauge setting 5.2 Tension setting 5.3 Pressure setting 5.4 Speed setting

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Alignment and calibration principles and tolerances 1.2 Shaft, belt and roller alignment methods 1.3 Calibration procedures for tension and pressure devices 1.4 Installation and commissioning requirements for textile machinery 1.5 Reading installation drawings and layout plans 1.6 Use and care of precision alignment and measuring instruments 1.7 Recording and reporting calibration results and deviations 1.8 Safety precautions during alignment, calibration and installation work
2. Underpinning Skills	2.1 Preparing for alignment, calibration and installation tasks 2.2 Carrying out shaft, belt and roller alignment 2.3 Calibrating machine settings, tension and pressure devices 2.4 Supporting installation and commissioning of new or relocated machines 2.5 Using precision alignment and measuring instruments correctly 2.6 Verifying alignment/calibration results through trial run and measurement 2.7 Interpreting installation drawings and layouts
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness

	3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Sample textile machine/drive assembly for alignment practice 4.3 Alignment and calibration tools (dial gauge, straight edge, feeler gauge, spirit level, tension meter, load, pressure & temperature calibrator) 4.4 Laser shaft alignment tool 4.5 Installation drawings/layouts 4.6 Calibration record/report formats 4.7 PPE set 4.8 Learning manual and projector

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 prepared for machine alignment, calibration and installation tasks 1.2 carried out shaft, belt and roller alignment 1.3 calibrated machine settings, tension and pressure devices 1.4 supported installation and commissioning of new or relocated machines
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated textile mill workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INTEGRATE CROSS-DISCIPLINARY TECHNOLOGY	Nominal Duration: 20 hrs.	Unit Code: SICIP-TEX-MMM-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to integrate cross-disciplinary technology relevant to modern textile machinery maintenance. It specifically includes the tasks of diagnosing and repairing integrated systems, repairing mechanical drives, interpreting PLC electrical circuits, and understanding advanced textile material sensors.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Diagnose and repair Integrated Systems	1.1 <u>Integrated mechanical-electrical systems</u> of modern textile machines are identified. 1.2 Faults involving both mechanical and electrical/control elements are recognized and referred/coordinated with electrical personnel as required. 1.3 Basic diagnostic checks on integrated systems are carried out within scope of authority. 1.4 Findings are coordinated with electrical/automation technicians for joint rectification.
2. Repair mechanical drives	2.1 <u>Mechanical drive systems</u> associated with motorized machine sections are identified. 2.2 Drive faults (slippage, misalignment, wear) are diagnosed and rectified. 2.3 Drive components are repaired/replaced and re-tensioned/aligned as required. 2.4 Drive performance is verified after repair.
3. Interpret PLC electrical circuits	3.1 Basic <u>PLC system components</u> are identified. 3.2 Simple PLC/electrical circuit diagrams related to machine operation are interpreted to understand machine sequence. 3.3 Input/output points relevant to mechanical systems (limit switches, proximity sensors, motor starters) are identified on the circuit diagram. 3.4 Observations from PLC/HMI fault indications are used to support mechanical fault diagnosis.
4. Understand advanced textile material sensors	4.1 <u>Sensors used in textile machinery</u> are identified according to their function. 4.2 Working principle of common sensors is explained in simple terms. 4.3 Sensor mounting position and mechanical adjustment affecting sensor performance are identified. 4.4 Sensor-related mechanical issues (misalignment, dirt, damage) are identified and addressed.

Range of Variables:

Variables	Range (may include but not limited to)
1. Integrated mechanical-electrical systems	1.1 Motor-drive assemblies 1.2 Sensor-actuator systems 1.3 PLC-controlled mechanical sections 1.4 Electro-pneumatic and electro-hydraulic
2. Mechanical drive systems	2.1 Belt drive 2.2 Chain drive 2.3 Gear drive 2.4 Direct coupling drive
3. PLC system components	3.1 CPU 3.2 Input modules 3.3 Output modules 3.4 Power supply unit 3.5 Communication module
4. Sensors used in textile machinery	4.1 Proximity sensor 4.2 Photoelectric sensor 4.3 Tension sensor 4.4 Speed sensor 4.5 Yarn/fabric break sensor 4.6 Temperature sensor 4.7 Pressure sensor

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Basic concept of integrated mechanical-electrical (mechatronic) systems 1.2 Mechanical drive systems and common drive faults 1.3 Basic PLC system components and function 1.4 Reading simple PLC/electrical circuit diagrams relevant to mechanical systems 1.5 Common sensors used in textile machinery and their working principle 1.6 Coordination procedures with electrical/automation personnel 1.7 Sensor mounting and mechanical adjustment factors affecting performance 1.8 Basic input/output points relevant to mechanical systems (limit switches, proximity sensors, motor starters)
2. Underpinning Skills	2.1 Identifying integrated mechanical-electrical systems and coordinating with electrical personnel 2.2 Diagnosing and repairing mechanical drive faults 2.3 Interpreting basic PLC electrical circuits relevant to mechanical systems 2.4 Identifying sensors and addressing sensor-related mechanical issues 2.5 Carrying out basic diagnostic checks on integrated systems within scope of authority 2.6 Using PLC/HMI fault indications to support mechanical fault diagnosis

	2.7 Re-tensioning and aligning drive components after repair
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Sample mechanical drive assembly (belt/chain/gear drive) 4.3 Basic PLC trainer kit and sample circuit diagrams 4.4 Sample sensors used in textile machinery (proximity, photoelectric, tension, etc.) 4.5 Sample PLC/HMI fault indication charts 4.6 Hand tools for drive repair 4.7 PPE set 4.8 Learning manual and projector

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 diagnosed and repaired integrated systems 1.2 repaired mechanical drives 1.3 interpreted PLC electrical circuits 1.4 demonstrated understanding of advanced textile material sensors
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated textile mill workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM BASIC WELDING, FABRICATION AND WORKSHOP SUPPORT OPERATIONS	Nominal Duration: 30 hrs.	Unit Code: SICIP-TEX-MMM-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform basic welding, fabrication and workshop support operations. It specifically includes the tasks of preparing for welding and fabrication tasks, carrying out basic welding, cutting and fabrication operations, fabricating or repairing simple machine parts, guards and fixtures, and maintaining workshop tools, machines and housekeeping.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for welding and fabrication tasks	1.1 Scope of welding/fabrication task is confirmed from job requirement. 1.2 <u>Welding and fabrication tools/equipment</u> are selected and checked for serviceability. 1.3 Materials (metal stock, electrodes, consumables) required are prepared. 1.4 PPE and safety measures (fire prevention, ventilation) for welding work are implemented.
2. Carry out basic welding, cutting and fabrication operations	2.1 <u>Basic welding processes</u> are carried out to join or repair simple metal parts. 2.2 Cutting operations (gas cutting/grinding) are carried out safely and accurately. 2.3 Fabricated joints/parts are checked for dimensional accuracy and quality. 2.4 Work is carried out following workshop safety procedures.
3. Fabricate or repair simple machine parts, guards and fixtures	3.1 Simple machine parts, brackets, guards or fixtures are fabricated according to drawing/requirement. 3.2 Damaged guards/fixtures are repaired to restore original function and safety. 3.3 Fabricated/repaired items are fitted and checked on the machine. 3.4 Finishing operations (grinding, deburring, painting) are carried out as required.
4. Maintain workshop tools, machines and housekeeping	4.1 Workshop machines and tools are checked, cleaned and maintained after use. 4.2 Tools and equipment are stored in designated locations following workshop procedure. 4.3 Workshop area is kept clean, organized and hazard-free (5S practice). 4.4 Faulty tools/equipment are reported and tagged for repair.

Range of Variables:

Variables	Range (may include but not limited to)
1. Welding and fabrication tools/equipment	1.1 Arc welding machine 1.2 Gas welding/cutting set 1.3 Grinder 1.4 Hand tools 1.5 Measuring/marketing tools
2. Basic welding processes	2.1 Shielded metal arc welding (SMAW) 2.2 Gas welding 2.3 Spot welding (where applicable)

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Basic welding processes and their application in machine repair 1.2 Cutting and fabrication techniques 1.3 Fabrication and repair of simple machine parts, guards and fixtures 1.4 Workshop tool maintenance and housekeeping (5S) practices 1.5 Selection and preparation of welding/fabrication tools and materials 1.6 Fire prevention and ventilation safety measures for welding work 1.7 Dimensional accuracy and quality checking of fabricated items 1.8 Reporting and tagging procedures for faulty tools/equipment
2. Underpinning Skills	2.1 Preparing for welding and fabrication tasks 2.2 Carrying out basic welding, cutting and fabrication operations 2.3 Fabricating or repairing simple machine parts, guards and fixtures 2.4 Maintaining workshop tools, machines and housekeeping 2.5 Selecting and checking welding/fabrication tools and equipment 2.6 Applying PPE and safety measures during welding operations 2.7 Checking dimensional accuracy and quality of fabricated/repaired items
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace/workshop (simulated or actual)

	4.2 Arc welding machine and gas welding/cutting set 4.3 Grinder and hand tools 4.4 Metal stock, electrodes and consumables 4.5 Sample guards/fixtures for fabrication/repair practice 4.6 Measuring/marketing tools 4.7 PPE set (welding shield, gloves, apron, etc.) 4.8 Learning manual and projector
--	---

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 prepared for welding and fabrication tasks 1.2 carried out basic welding, cutting and fabrication operations 1.3 fabricated or repaired simple machine parts, guards and fixtures 1.4 maintained workshop tools, machines and housekeeping
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated textile mill workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MAINTAIN MAINTENANCE RECORDS, SPARE PARTS INVENTORY AND SAFETY COMPLIANCE	Nominal Duration: 30 hrs.	Unit Code: SICIP-TEX-MMM-08-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to maintain maintenance records, spare parts inventory and safety compliance. It specifically includes the tasks of maintaining machine history cards and maintenance logs, managing spare parts inventory and requisition process, monitoring compliance with safety, environmental and housekeeping standards, and reporting equipment downtime, breakdowns and maintenance KPIs.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Maintain machine history cards and maintenance logs	1.1 <u>Machine history card</u> is updated with all maintenance, repair and breakdown events. 1.2 Maintenance logs (daily/shift) are completed accurately and on time. 1.3 Records are maintained in a legible and organized manner (manual or digital). 1.4 Records are made available for audit/review when required.
2. Manage spare parts inventory and requisition process	2.1 <u>Spare parts inventory</u> is checked and stock levels are monitored against minimum/maximum limits. 2.2 Spare parts requisition is raised following organizational procurement procedure. 2.3 Received spare parts are checked, recorded and stored appropriately. 2.4 Critical/slow-moving spares are identified and reported for inventory planning.
3. Monitor compliance with safety, environmental and housekeeping standards	3.1 <u>Workplace safety and environmental requirements</u> applicable to maintenance work are identified. 3.2 Compliance with safety procedures (PPE use, lockout-tagout, waste disposal) is monitored. 3.3 Housekeeping (5S) standards are maintained in work and storage areas. 3.4 Non-compliance issues are reported to the supervisor for corrective action.
4. Report equipment downtime, breakdowns and maintenance KPIs	4.1 Equipment downtime and breakdown events are recorded accurately with cause and duration. 4.2 <u>Maintenance KPIs</u> are calculated/compiled as required. 4.3 Reports are prepared and submitted to supervisor/management according to schedule. 4.4 Trends and recurring issues identified from reports are highlighted for improvement action.

Range of Variables:

Variables	Range (may include but not limited to)
1. Machine history card	1.1 Manual record card

	1.2 Digital/CMMS record
2. Spare parts inventory	2.1 Critical spares 2.2 Fast-moving spares 2.3 Slow-moving spares 2.4 Minimum-maximum stock levels
3. Workplace safety and environmental requirements	3.1 PPE requirements 3.2 Waste disposal procedure 3.3 Fire safety 3.4 Noise/emission control
4. Maintenance KPIs	4.1 MTBF (Mean Time Between Failures) 4.2 MTTR (Mean Time To Repair) 4.3 PM compliance rate 4.4 Maintenance cost per unit

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Machine history card and maintenance log procedures 1.2 Spare parts inventory management and requisition process 1.3 Workplace safety, environmental and housekeeping (5S) standards 1.4 Maintenance KPIs and their calculation (MTBF, MTTR, PM compliance) 1.5 Digital/manual record-keeping systems 1.6 Procurement and vendor coordination procedures for spare parts 1.7 Identification of critical and slow-moving spare parts 1.8 Downtime and breakdown reporting procedures
2. Underpinning Skills	2.1 Maintaining machine history cards and maintenance logs 2.2 Managing spare parts inventory and the requisition process 2.3 Monitoring compliance with safety, environmental and housekeeping standards 2.4 Reporting equipment downtime, breakdowns and maintenance KPIs 2.5 Checking and recording received spare parts accurately 2.6 Identifying and reporting non-compliance issues to the supervisor 2.7 Calculating and compiling maintenance KPIs (MTBF, MTTR, PM compliance)
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Machine history card and maintenance log formats 4.3 Spare parts inventory and requisition formats

	4.4 Sample spare parts bin cards/stock registers 4.5 Safety/environmental compliance checklist 4.6 Sample downtime and KPI report formats 4.7 Waste segregation bins and disposal procedure reference 4.8 Learning manual and projector
--	---

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 maintained machine history cards and maintenance logs 1.2 managed spare parts inventory and requisition process 1.3 monitored compliance with safety, environmental and housekeeping standards 1.4 reported equipment downtime, breakdowns and maintenance KPIs
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated textile mill workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

WORKSHOP / LAB FACILITY STANDARD

Course Name:	Machine Management and Maintenance for Textile Industry
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

1. Classroom - 400 sft (37 sqm)
2. Workshop/Lab - 2000 sft (186 sqm), **OR**
3. Classroom cum workshop - 2300 sft (214 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training Facilities	Required Quantity
1.	Computer/Laptop	1
2.	Multimedia projector & screen	1
3.	Printer	1
4.	Scanner	1
5.	Internet connection with Wi-Fi facility	1
6.	Whiteboard with accessories	1
7.	Sample textile machine sub-assemblies/cutaway models (spinning, weaving, knitting)	1 set
8.	Machine manuals, drawings and maintenance schedule samples	1 set
9.	PPE set (safety helmet, safety shoes, gloves, eye/face protection, ear protection, dust mask)	25 sets
10.	Preventive maintenance checklist and machine history card formats (sample kit)	1 set
11.	Grease gun (manual/pneumatic)	5
12.	Oil can/dispenser	5
13.	Assorted lubricants and greases (training stock)	1 lot
14.	Vibration pen/meter	2
15.	Infrared thermometer/thermal imaging camera	2
16.	Mechanic's stethoscope/listening rod	5
17.	Tachometer (contact/non-contact)	2
18.	Clearance/feeler gauge set	5 sets
19.	Bearing puller set (mechanical/hydraulic)	3 sets
20.	Bearing heater/induction heater	1
21.	Assorted bearings, gears, belts, chains and shafts (training stock)	1 lot
22.	Vernier caliper and micrometer set	5 sets
23.	Torque wrench set	3 sets
24.	Hand tool set (spanner, screwdriver, plier, nipper, Allen key)	5 sets
25.	Hydraulic/pneumatic puller	2
26.	Chain block/hoist with lifting tackle	1
27.	Lifting jack	2
28.	Dial gauge with magnetic stand	5
29.	Laser shaft alignment tool	1

30.	Straight edge and spirit level set	5 sets
31.	Tension meter	2
32.	Motor-coupling/drive alignment training rig	2
33.	Basic PLC trainer kit with I/O modules	2
34.	Sample sensors (proximity, photoelectric, tension) with mounting fixtures	1 set
35.	Mechanical drive training rig (belt/chain/gear drive)	2
36.	Arc welding machine	2
37.	Gas welding/cutting set	1 set
38.	Bench/pedestal grinder	2
39.	Welding workbench with vice	4
40.	Welding PPE (face shield, gloves, apron)	10 sets may be 25 sets
41.	Assorted metal stock, electrodes and welding consumables	1 lot
42.	Spare parts inventory/bin-card training kit	1 set
43.	Lockout-tagout (LOTO) kit	2 sets
44.	Material handling trolley	2
45.	Industrial vacuum cleaner/air blowing gun	2
46.	Waste segregation bins (colour coded)	4
47.	Tool board and storage cabinet	3
48.	Fire extinguisher	2
49.	First aid box	1
50.	Some metals/alloy (Cu, Al, CI, Steel etc.)	1 lot

The following conditions must be fulfilled –

1. The institute shall not use the same facilities for any other projects/organizations offering a similar course.
2. The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facilities required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirements of the Competency Standard (CS) provided by SICIP.

For the operation of the training course on Machine Management and Maintenance (Mechanical) for Textile Industry, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

Where a training institute does not possess the full range of production and maintenance machinery, the institute must establish a formal linkage with a spinning, weaving and/or knitting mill's maintenance department so that the practical sessions of the occupation-specific units can be conducted on actual production machinery under supervision.

The institute will also provide all other hand tools and power tools as per the CS for 25 trainees. The institute will also arrange adequate seating arrangements and classroom setup for the 25 trainees.